

Elements of Statistics and Probability

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1 Correlation

Why Correlation?

- There are many situations where the objective of the study is to see the relationship among variables. Consider the following situations:
 - Is there any relation between shoe size and height?
 - Is there any relation between number of hours you exercise every day and your heart rate?
 - Is there any relation between math scores and overall scores in exam?
 - Is there any relation between temperature and earthquake?

Correlation

- It is concerned with **two** (or more) sets of data, **not** with only **one** set.
- When we measure interrelationship between
 - **two** variables, then it is called **simple** correlation
 - among **more than two** variables is termed as **multiple** correlation
- **Definition:** The **correlation** is the measure of the extent and the direction of the relationship between two variables in a bivariate distribution

Advantages of Correlation Analysis

- Correlation analysis measures the strength of relationship between the variables
- It also provides the direction of the relationship between the variables
- It reduces the range of uncertainty in matter of prediction; prediction is not perfectly accurate here, but helpful

Disadvantages of Correlation Analysis

- Correlation only describes a relationship. It does not tell us why
- Correlation is affected greatly by the range of the data set
- If there is outlier in the dataset, it significantly alters the correlation
- Correlation does not capture strong non-linear relationship between variables

Types of Correlation

- **Positive correlation:** Variables change in the same direction
 - If X increases, Y also increases
 - If X decreases, Y also decreases
- **Negative correlation:** Variables change in the opposite direction
 - If X increases, Y decreases
 - If X decreases, Y increases

Examples

- Income and expenditure on luxury goods
- Weekly study hour and CGPA of students
- Spending on advertisement and sales
- Price and demand of a commodity
- Class attendance and quiz marks

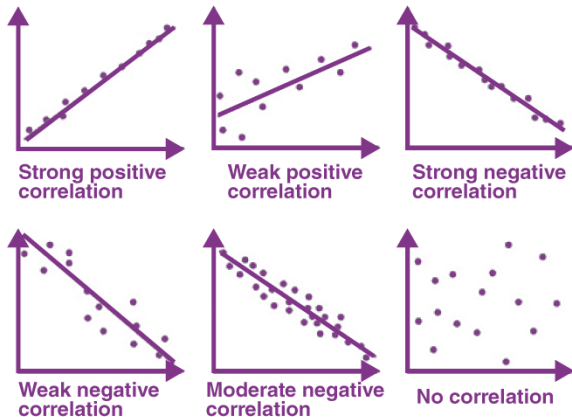
How to Determine Correlation?

- Graphical Methods:
 - ① Scatter diagram
- Algebraic methods:
 - ① Karl Pearson's coefficient of correlation
 - ② Spearman's rank correlation

Scatter Diagram to Visualise Correlation

- It is the simplest device for ascertaining if the two variables are related or not
- The given data are plotted on a graph paper in the form of dots
- The **greater** the scatter of points over the graph, the **lesser** is the relationship between the variables

Scatter Diagram to Visualise Correlation (cont.)



Pros and Cons of a Scatter Diagram

- Merits:

- Simple and non-mathematical method
- Easily understandable
- Not influenced by extreme values

- Demerits:

- Can't establish the exact degree of correlation between the variables

Karl Pearson's Correlation Coefficient

- It is also known as Pearson's product-moment correlation coefficient
- This is the most popular measure of correlation for measuring the degree or strength of linear relationship between two variables
- Assumptions:
 - 1 The underlying relationship between two variables is linear, which is the assumption of linearity
 - 2 The variables under consideration should have a bivariate normal distribution

Karl Pearson's Correlation Coefficient (cont.)

- Karl Pearson's Correlation Coefficient:

$$\begin{aligned} r &= \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^n (x_i - \bar{x})^2 \sum_{i=1}^n (y_i - \bar{y})^2}} \\ &= \frac{\sum_{i=1}^n x_i y_i - n\bar{x}\bar{y}}{\sqrt{(\sum_{i=1}^n x_i^2 - n\bar{x}^2)(\sum_{i=1}^n y_i^2 - n\bar{y}^2)}} \\ &= \frac{\text{cov}(x, y)}{s_x s_y} \end{aligned}$$

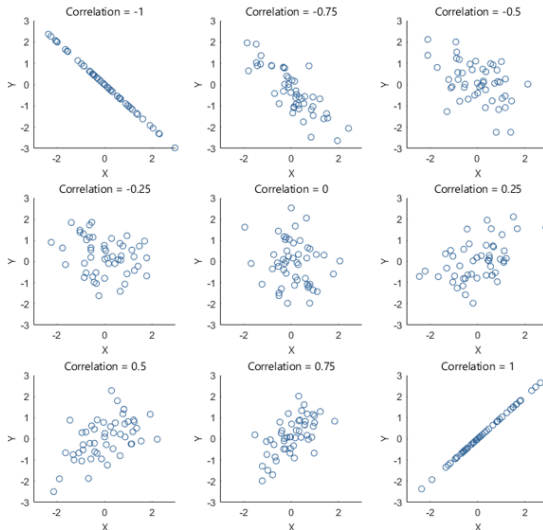
- where $\text{cov}(x, y) = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{n - 1}$ is the sample covariance, and s_x & s_y are the sample standard deviations for X and Y

Karl Pearson's Correlation Coefficient (cont.)

- Interpretations of r :

Degree of association	Value of r	
	Positive	Negative
Perfect	+1	-1
High	$+0.75 \leq r < +1$	$-1 < r \leq -0.75$
Moderate	$+0.25 \leq r < +0.75$	$-0.75 < r \leq -0.25$
Low	$0 < r < +0.25$	$-0.25 < r < 0$
No correlation	0	0

Karl Pearson's Correlation Coefficient (cont.)



Pros and Cons of r

- Merits:

- It summarizes in one value the degree of correlation and also the direction of the correlation
- It gives us exact measure of degree of association
- It is independent of units of the variables

- Demerits:

- Value of r is affected by extreme values
- Calculation method is long, tedious and time consuming
- Only applicable for **quantitative** measurements
- When the population is **not normally distributed**, r may underestimate the relationship between the variables

Coefficient of Determination

- The coefficient of determination is the square of the coefficient of correlation r , that is r^2
- It is the measure of the strength of the relationship between two variables
- It measures the **percentage of variation** of one variable which can be described by another variable
- The value of determination coefficient lies between 0 and 1
- Suppose $r = 0.9$, then $r^2 = 0.81$. This would mean that 81% of total variation in y can be explained by x

Example

- A small experiment is conducted involving 10 students to investigate the association between students' math scores and overall scores in exam.

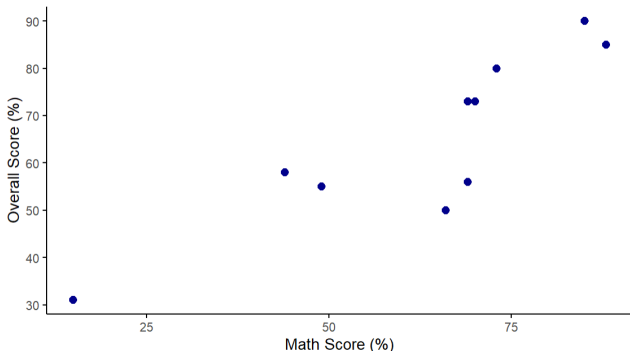
Math score (%):	70	85	66	15	69	49	73	44	88	69
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Overall score (%):	73	90	50	31	56	55	80	58	85	73
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- Is there any linear association between math score and overall score?

Example (cont.)

- Scatter plot:



- Comment:** This graph shows positive correlation between math scores and overall scores

Example (cont.)

x	y	xy	x^2	y^2
70	73	5110	4900	5329
85	90	7650	7225	8100
66	50	3300	4356	2500
15	31	465	225	961
69	56	3864	4761	3136
49	55	2695	2401	3025
73	80	5840	5329	6400
44	58	2552	1936	3364
88	85	7480	7744	7225
69	73	5037	4761	5329

$$\Sigma x_i = 628 \quad \Sigma y_i = 651 \quad \Sigma x_i y_i = 43993 \quad \Sigma x_i^2 = 43638 \quad \Sigma y_i^2 = 45369$$

- $\bar{x} = \Sigma x_i / n = 628 / 10 = 62.8, \quad \bar{y} = \Sigma y_i / n = 651 / 10 = 65.1$

Example (cont.)

- Correlation coefficient:

$$\begin{aligned} r &= \frac{\sum_{i=1}^n x_i y_i - n\bar{x}\bar{y}}{\sqrt{(\sum_{i=1}^n x_i^2 - n\bar{x}^2)(\sum_{i=1}^n y_i^2 - n\bar{y}^2)}} \\ &= \frac{43993 - (10 \times 62.8 \times 65.1)}{\sqrt{(43638 - 10 \times 62.8^2)(45369 - 10 \times 65.1^2)}} \\ &= 0.878 \end{aligned}$$

- Coefficient of determination: $r^2 = (0.878)^2 = 0.771$

Example (cont.)

- Interpretation of $r = 0.878$:
 - There exists high degree of positive correlation between math scores (x) and overall scores (y) of students
- Interpretation of $r^2 = 0.771$:
 - 77.1% of total variation in overall score (y) can be explained by the linear relationship between math scores (x) and overall scores (y)
 - The other 22.9% of the total variation in overall score (y) remains unexplained

Exercise 1

- A small business wants to explore whether its spending on online advertising affects its monthly sales revenue. Data for the last 10 months are shown below:

Advertising Cost ('000 BDT):	25	30	20	15	28	32	35	22	40	18
Sales Revenue ('000 BDT):	220	240	180	150	230	250	270	200	290	160

- Draw a scatterplot of Sales Revenue against Advertising Cost. Describe the type and strength of relationship you observe.
- Calculate and comment on the Pearson correlation coefficient (r) between Advertising Cost and Sales Revenue.
- Compute the coefficient of determination. Interpret this value

Exercise 2

- A biologist recorded the activity level of a certain enzyme at different temperatures to understand how heat affects reaction efficiency. The results are shown below:

Temperature ($^{\circ}\text{C}$):	20	25	30	35	40	45	50	55	60	65
Enzyme Activity ($\mu\text{ mol/min}$):	22	28	35	42	48	52	51	43	33	20

- Draw a scatterplot of Enzyme Activity against Temperature. Comment on it.
- Calculate the Pearson correlation coefficient. Interpret the result.
- Calculate the r^2 . Discuss what proportion of the change in Enzyme Activity is explained by Temperature.

Thank You